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Non-mainstream behaviors as endogenous perturbations in high-density public spaces: Evidence, modeling, and interventions

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ABSTRACT

Pedestrian movement in high-density public spaces is essential for safe and efficient urban mobility, yet the system-level effects of movement heterogeneity remain insufficiently understood. Here, we develop a data-driven framework combining empirical trajectory analysis with an adapted social force model to investigate how non-mainstream behaviors affect pedestrian flows. A segment-based spatial cell method consistently identifies these behavioral deviations across real-world contexts. Empirical results show that non-mainstream behaviors increase local speed variability, reduce local flow, and are associated with greater congestion levels. Simulations reveal synergistic effects of crowd density and non-mainstream behavior: increasing density intensifies compression, while higher non-mainstream ratios introduce additional friction and weaken collective order, jointly amplifying congestion and crowd danger. Geometric interventions reorganize heterogeneous flows, mitigate conflicts, and improve efficiency and safety. These findings highlight the role of movement heterogeneity in crowd instability and inform spatial interventions for safer and more resilient pedestrian built environments.

1. Introduction

Pedestrian flows in high-density public spaces (Hou et al., 2023), such as transit hubs, stadium corridors, and open squares, are fundamental to the operational efficiency and safety of urban infrastructure (Zhao et al., 2018). In these environments, pedestrians often form macroscopic unidirectional flows driven by structural layouts and collective navigation goals. Understanding the capacity and stability of pedestrian flows is critical, as minor disruptions in dense crowds may rapidly escalate into localized congestion (Wu et al., 2019), stop-and-go waves (Moussaïd et al., 2011), or even severe crowd stampede incidents (Sharma et al., 2023). Therefore, uncovering the underlying mechanisms that drive flow instability remains a central focus in pedestrian dynamics and crowd safety management (Sharma et al., 2016).

In recent years, extensive empirical and modeling studies have explored how localized structural perturbations influence pedestrian flow dynamics. Empirical studies show that obstacles can alter pedestrian outflow, congestion, lane formation, and evacuation efficiency, with their effects depending strongly on obstacle size, position, and geometric arrangement (Garcimartín et al., 2018; Shi et al., 2019; Chen et al., 2019; Zhao et al., 2020; Jia et al., 2021; Ding et al., 2025). Modeling and simulation studies further suggest that structural elements can reorganize pedestrian trajectories, reduce high-density regions, regulate local interactions, and improve flow efficiency, while their effectiveness remains sensitive to obstacle geometry and placement (Jiang et al., 2014; Jia et al., 2017; Zhao et al., 2017; Sticco et al., 2022; Qiu et al., 2025; Wu et al., 2025; Zhou et al., 2025). Collectively, these studies establish that localized structural perturbations can substantially reshape pedestrian interactions and collective flow patterns (Shiwakoti et al., 2019). However, they primarily consider static structural elements, while pedestrians themselves may also act as dynamic or quasi-static sources of local disturbance. In real-world crowds, variations in individual speed, movement direction, and short-term behavioral deviations (Wu et al., 2022a,b) can influence congestion, flow stability, and collective motion. This motivates extending the perturbation perspective from external structural obstacles to internally generated pedestrian heterogeneity.

From this perspective, heterogeneous pedestrian flows exhibit distinct collective characteristics arising from differences in movement direction, speed, and flow composition. Studies of bidirectional and counter-flow conditions show that directional composition, density, and local flow conditions can substantially influence lane formation, phase transitions, and flow stability (Feliciani and Nishinari, 2016; Liu et al., 2024). Multidirectional merging flows likewise exhibit complex interaction patterns that vary with flow level and spatial organization

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(Yu et al., 2025). In parallel, trajectory-based studies have characterized heterogeneous pedestrian states and movement behaviors using combinations of walking speed, movement direction, occupancy, and kinematic features together with clustering and related data-driven approaches (Masumura and Sadohara, 2023; McArdle et al., 2014). Specific movement patterns, such as pedestrian crossing behaviors, have also been identified from trajectory features including walking speed and path deviation using clustering-based methods (Fu et al., 2023). Together, these studies demonstrate that pedestrian heterogeneity can be characterized through multiple kinematic and spatial features. Nevertheless, most existing approaches focus on predefined flow types or aggregate movement patterns, while transient non-mainstream deviations relative to the locally prevailing flow remain less systematically characterized.

Building on this, a limited but growing body of work has begun to explicitly treat pedestrians as endogenous perturbations in collective flows. Empirically, Moussaïd et al. (2012) conducted laboratory experiments on bidirectional flows, demonstrating that local interactions between faster and slower pedestrians induce traffic instabilities. Ma et al. (2013) analyzed video recordings before the Love Parade disaster and revealed that stopped pedestrians in dense counter-flow act as internal perturbations, triggering force transmission and velocity fluctuations that lead to turbulent-like motion. Hu et al. (2023) introduced a stationary pedestrian as a localized obstacle and observed reduced entry speeds but alleviated crowd pressure during the inflow process, while their subsequent study (Hu et al., 2025) further showed that such human-induced obstructions at bottlenecks can deteriorate evacuation performance by increasing escape time and time headway. On the modeling side, Yang et al. (2024) improved the social force model to study collision avoidance in counter-flow, suggesting that strong avoidance behaviors hinder lane formation, whereas higher flow ratios promote it. Similarly, Li et al. (2025) developed a geometric cognition model and discovered that more aggressive behaviors induce stop-and-go waves, weaken lane structures, and disrupt overall stability.

Overall, these studies reflect that movement heterogeneity among pedestrians has varying levels of influence on flow organization and both efficiency and stability. However, current research remains limited in several respects. On the one hand, existing work typically focuses on isolated behavioral factors (e.g., stopping, avoidance, or speed heterogeneity) rather than providing a unified and comprehensive quantification of non-mainstream behaviors in real-world pedestrian flows. On the other hand, the extent to which such microscopic behavioral deviations translate into macroscopic flow instability, particularly under different crowd conditions (e.g., density levels and behavioral compositions), remains insufficiently understood. The above limitations point to an important unresolved question: how can we systematically characterize non-mainstream behaviors and clarify their role in affecting the stability and performance of pedestrian flows?

To address the question, this work proposes a comprehensive framework combining empirical trajectory analysis and numerical modeling to investigate the effects of non-mainstream behaviors on pedestrian flow dynamics. First, we propose a segment-based spatial cell approach to identify non-mainstream behaviors from real-world datasets, and quantify their localized microscopic effects on fundamental traffic quantities. Second, we develop an adapted social force model to systematically scan the macroscopic impacts of varying crowd densities and non-mainstream ratios on kinematic and safety metrics. Finally, motivated by the need for crowd flow regulation, we introduce and evaluate structural control strategies, demonstrating their effectiveness in disentangling conflicting trajectories, promoting traffic efficiency, and alleviating crowd congestion. These findings advance the understanding of behavior-induced pedestrian flow instabilities and provide actionable insights for the management of efficient and safe crowd environments.

This paper is organized as follows. Section 2 describes the empirical data analysis for non-mainstream behaviors and the adapted social force model for numerical simulations. Section 3 presents the identification of non-mainstream behaviors, their quantified microscopic and macroscopic impacts, and the evaluation of structural control strategies. Last, main conclusions, underlying implications, and future prospects are discussed in Section 4.

2. Methods

2.1. Empirical data analysis: non-mainstream behaviors

2.1.1. Dataset description

The empirical analysis is based on the LargeView datasets, which were collected during the 2022 Festival of Lights in Lyon, France (Dufour et al., 2025). This event was a major urban gathering and attracted more than two million visitors over four evenings, with extremely dense pedestrian crowds forming around key locations such as Place des Terreaux, where nightly attendance reached 150,000 individuals. Such conditions provided a rare opportunity to observe pedestrian dynamics in highly dense and heterogeneous real-world environments, with local densities reaching up to $4\,\mathrm{ped/m^2}$. The video footage of the LargeView datasets was obtained from a camera installed on the City Hall tower at approximately 48 m above ground level, providing a bird's-eye view of the pedestrian flows at Place des Terreaux. In this study, we focus on two temporal subsets, namely LargeView_Zoom_A and LargeView_Zoom_O. Both were extracted from the same fixed 15 m × 25 m observation area shown in Fig. 1, located near a crucial high-density convergence point, but corresponded to two distinct periodic egress episodes recorded at different times. The trajectories of all observable people in this area were semi-manually extracted and upsampled to 10 FPS by linear interpolation. These datasets capture complex, non-uniform flow patterns, offering a suitable empirical basis for analyzing non-mainstream behaviors in real-world environments.

2.1.2. Identifying non-mainstream behaviors

The measurement area A is partitioned into a regular grid composed of cells with side length $a = 1$ m. The pedestrian trajectory L_i is decomposed into a set of continuous segments $L_i = \{l_i^1, l_i^2, \ldots, l_i^n\}$, each confined to a single grid cell. The kth segment l_i^k corresponds to a sequence of position vectors at consecutive time instants:

$$l_i^k = \left\{ \mathbf{x}_i(t) \mid t \in \mathcal{T}_i^k \right\} \tag{1}$$

where $\mathbf{x}_i(t)$ is the position of pedestrian i at time t, and $\mathcal{T}_i^k$ denotes a set of consecutive time indices, with cardinality $T_i^k = |\mathcal{T}_i^k|$. From this, the segment-average speed v_i^k is defined as:

$$v_i^k = \frac{1}{T_i^k} \sum_{t \in \mathcal{T}_i^k} \|\mathbf{v}_i(t)\| \tag{2}$$

Here, the instantaneous velocity $\mathbf{v}_i(t) = \left[\mathbf{x}_i(t + \Delta t) - \mathbf{x}_i(t)\right] / \Delta t$ is derived using a forward finite-difference method, where $\Delta t = 0.1$ s is the time interval between consecutive frames. The segment-average direction is computed in a similar way:

$$\mathbf{e}_i^k = \frac{1}{\hat{T}_i^k} \sum_{t \in \hat{\mathcal{T}}_i^k} \frac{\mathbf{v}_i(t)}{\|\mathbf{v}_i(t)\|} \tag{3}$$

where $\hat{\mathcal{T}}_i^k = \{t \in \mathcal{T}_i^k \mid \|\mathbf{v}_i(t)\| > 0\}$ denotes the set of time instants in the segment with non-zero speed, with cardinality $\hat{T}_i^k = |\hat{\mathcal{T}}_i^k|$. Together, v_i^k and $\mathbf{e}_i^k$ provide a temporally smoothed and spatially localized characterization of pedestrian motion within each grid cell.

For each cell $c(p, q)$, all trajectory segments passing through that cell are averaged to calculate the cell-level mainstream direction:

$$\mathbf{e}_{c(p,q)} = \frac{1}{N_{c(p,q)}^{\mathrm{seg}}} \sum_{l_i^k \in c(p,q)} \frac{\mathbf{e}_i^k}{\|\mathbf{e}_i^k\|} \tag{4}$$

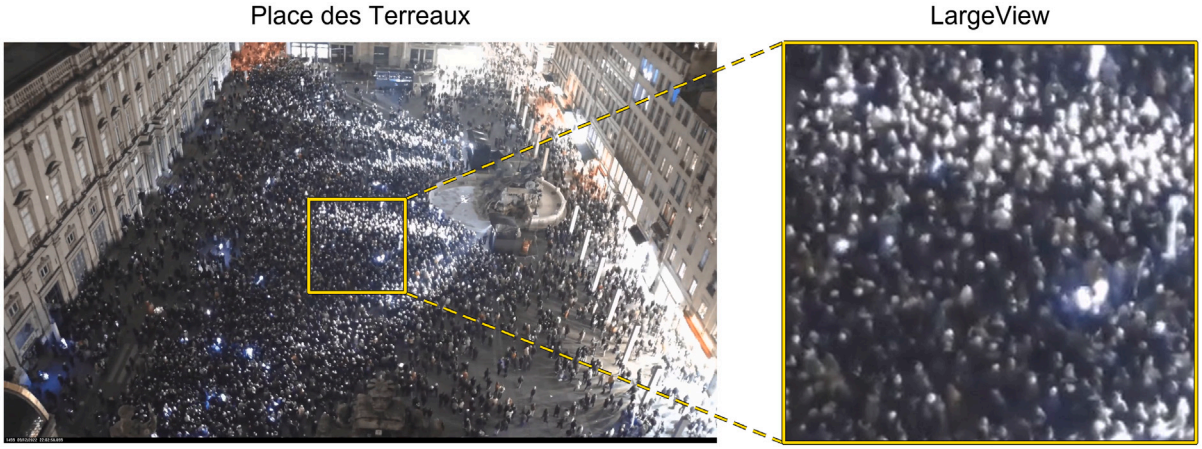


Fig. 1. Overview of Place des Terreaux and the LargeView observation area. The yellow box in the left panel marks the common 15 m × 25 m tracking region for LargeView_Zoom_A and LargeView_Zoom_O. An enlarged view of this region is shown on the right.

where $N^{\text{seg}}_{c(p,q)}$ represents the number of segments in cell $c(p,q)$, and only cells with sufficient sample size ($N^{\text{seg}}_{c(p,q)} \geq N^{\text{seg}}_{\min} = 5$) are considered for analysis.

Here, we define a trajectory segment l^k_i in cell $c(p,q)$ to be categorized as mainstream if and only if it satisfies two requirements: (1) its average speed v^k_i exceeds a minimal threshold v^*; and (2) its directional deviation $\Delta\theta_{ic}$ from the cell-level mainstream direction is below a prescribed tolerance θ^*, with the directional deviation defined as:

$$\Delta\theta_{ic} = \arccos\left(\frac{\mathbf{e}^k_i}{\|\mathbf{e}^k_i\|} \cdot \frac{\mathbf{e}_{c(p,q)}}{\|\mathbf{e}_{c(p,q)}\|}\right) \tag{5}$$

while segments failing either condition are classified as non-mainstream. For the speed threshold, pedestrian studies have used different values to distinguish stationary behavior from normal walking, often reflecting study-specific assumptions or conventions (Moussaïd et al., 2011; Ziemer et al., 2016; Zeng et al., 2019). Here, we adopt $v^* = 0.2$ m/s, informed by our recent analysis of more than 20 million pedestrian trajectories (Wu et al., 2026). The directional threshold $\theta^* = 45°$ follows a symmetric partition into four equal 90° sectors centered on the local forward, backward, and two transverse directions, providing an interpretable geometric distinction between predominantly forward movement and inverse or crossing movement.

In addition, two cell-level indicators are introduced to characterize the local motion dynamics within each grid cell. First, the cell segment frequency is defined as:

$$f_{c(p,q)} = \frac{N^{\text{seg}}_{c(p,q)}}{T_{\text{tot}}} \tag{6}$$

where $N^{\text{seg}}_{c(p,q)}$ represents the number of trajectory segments occurring in cell $c(p,q)$, and T_{tot} denotes the total observation duration (73.3 s for LargeView_Zoom_A and 119.5 s for LargeView_Zoom_O). This metric can be interpreted as a temporal flux-like quantity, reflecting how frequently pedestrians pass through the cell. Second, the cell non-mainstream ratio is measured as:

$$r_{c(p,q)} = \frac{N^{\text{non}}_{c(p,q)}}{N^{\text{seg}}_{c(p,q)}} \tag{7}$$

where $N^{\text{non}}_{c(p,q)}$ is the number of trajectory segments classified as non-mainstream in cell $c(p,q)$. This ratio quantifies the degree to which pedestrian behavior deviates from the local mainstream pattern.

The sensitivity analysis is conducted by independently varying a, v^*, and θ^* by ±20% in 10% increments while holding other parameters at their baseline values. Across both datasets, the resulting identification patterns and the principal relationships in the subsequent analyses remain qualitatively consistent across the tested settings. These results indicate that the main findings are not driven by the exact choices of the three parameters and are robust to moderate parameter variations.

2.1.3. Microscopic analytical metrics

To quantitatively characterize the local crowd dynamics around mainstream and non-mainstream pedestrians, we adopt a set of microscopic metrics based on the kernel function approach (Helbing et al., 2007), which enables a continuous description of local fields by weighting neighbors according to their spatial proximity.

The local density at position $\mathbf{x}$ and time t is computed as a weighted sum over all neighboring pedestrians i:

$$\rho(\mathbf{x},t) = \sum_i f\left(d_{i\mathbf{x}},t\right) \tag{8}$$

where $d_{i\mathbf{x}} = \|\mathbf{x}_i - \mathbf{x}\|$ is the distance between position $\mathbf{x}$ and pedestrian i, and $f(d)$ is a distance-dependent Gaussian kernel function:

$$f(d) = \frac{1}{\pi R^2}\exp\left(-\frac{d^2}{R^2}\right) \tag{9}$$

where R is a measurement parameter controlling the spatial influence range, and the value $R = 1$ m is a frequently used estimate.

The local speed variability is defined to capture the level of velocity fluctuations around position $\mathbf{x}$, reflecting local disorder and instability (Yi et al., 2024; Yi and Wu, 2026):

$$v(\mathbf{x},t) = \frac{1}{V(\mathbf{x},t)}\sqrt{\frac{\sum_i \left\|\mathbf{V}(\mathbf{x},t) - \mathbf{v}_i(t)\right\|^2 f\left(d_{i\mathbf{x}},t\right)}{\sum_i f\left(d_{i\mathbf{x}},t\right)}} \tag{10}$$

where the local speed is $V(\mathbf{x},t) = \|\mathbf{V}(\mathbf{x},t)\|$, and the velocity vector is calculated as the weighted average:

$$\mathbf{V}(\mathbf{x},t) = \frac{\sum_i \mathbf{v}_i(t) f\left(d_{i\mathbf{x}},t\right)}{\sum_i f\left(d_{i\mathbf{x}},t\right)} \tag{11}$$

According to the fluid-dynamic formula, the local flow is determined as local density multiplied by local speed:

$$Q(\mathbf{x},t) = \rho(\mathbf{x},t)\, V(\mathbf{x},t) \tag{12}$$

This metric describes the effective pedestrian capacity at the microscopic level.

2.2. Modeling framework: adapted social force model

To reproduce and further investigate pedestrian dynamics observed in real-world datasets, we use the social force model proposed by Helbing et al. (2000) for numerical simulations. This model has been

widely validated in describing pedestrian motion in open spaces (Wu and Yi, 2026b,a), where individual movement is governed by a combination of self-driven motivation and interactions with surrounding pedestrians and boundaries. In this study, we retain the basic force formulation of the social force model while adapting its behavioral and simulation settings, including desired speed, desired direction, and pedestrian generation rules, to characterize mainstream and non-mainstream movement behaviors.

The position $\mathbf{x}_i(t)$ of pedestrian i is updated by:

$$\frac{d\mathbf{x}_i(t)}{dt} = \mathbf{v}_i(t) \tag{13}$$

where the change of velocity $\mathbf{v}_i(t)$ is governed by the following acceleration equation:

$$m_i \frac{d\mathbf{v}_i(t)}{dt} = \mathbf{f}_{id} + \sum_{j(\neq i)} \mathbf{f}_{ij} + \sum_{W} \mathbf{f}_{iW} \tag{14}$$

The first term $\mathbf{f}_{id}$ represents a self-driven force that propels pedestrians toward the destination:

$$\mathbf{f}_{id} = m_i \frac{v_i^0(t)\,\mathbf{e}_i^0(t) - \mathbf{v}_i(t)}{\tau_i} \tag{15}$$

where pedestrian i of mass m_i tends to adapt his/her actual velocity $\mathbf{v}_i(t)$ within a characteristic time τ_i, so as to move with a desired speed $v_i^0(t)$ in a desired direction $\mathbf{e}_i^0(t)$.

The second term $\mathbf{f}_{ij}$ accounts for interactions with other pedestrians j:

$$\mathbf{f}_{ij} = A_i \exp\left[\left(r_{ij} - d_{ij}\right)/B_i\right]\mathbf{n}_{ij} + kg\left(r_{ij} - d_{ij}\right)\mathbf{n}_{ij} + \kappa g\left(r_{ij} - d_{ij}\right)\Delta v_{ji}^t \mathbf{t}_{ij} \tag{16}$$

Here, the psychological tendency of pedestrians i and j to stay away from each other is described by a repulsive interaction force $A_i \exp\left[\left(r_{ij} - d_{ij}\right)/B_i\right]\mathbf{n}_{ij}$, where A_i and B_i are constants, r_{ij} is the sum of their radii r_i and r_j, and $\mathbf{n}_{ij}$ denotes the normalized vector pointing from pedestrian j to i. Two additional forces, a body force $kg\left(r_{ij} - d_{ij}\right)\mathbf{n}_{ij}$ and a sliding friction force $\kappa g\left(r_{ij} - d_{ij}\right)\Delta v_{ji}^t \mathbf{t}_{ij}$, are introduced to counteract body compression and impede relative tangential motion, respectively. The parameters k and κ reflect the obstruction effects caused by physical contact, $\mathbf{t}_{ij} = \left(-n_{ij}^2, n_{ij}^1\right)$ is the tangential direction, and $\Delta v_{ji}^t = \left(\mathbf{v}_j - \mathbf{v}_i\right)\cdot\mathbf{t}_{ij}$ is the tangential velocity difference. The function $g(x)$ is zero if pedestrians i and j do not touch each other; otherwise, it is equal to the argument x.

The third term $\mathbf{f}_{iW}$ considers interactions between pedestrian i and wall W:

$$\mathbf{f}_{iW} = A_i \exp\left[\left(r_i - d_{iW}\right)/B_i\right]\mathbf{n}_{iW} + kg\left(r_i - d_{iW}\right)\mathbf{n}_{iW} - \kappa g\left(r_i - d_{iW}\right)\left(\mathbf{v}_i \cdot \mathbf{t}_{iW}\right)\mathbf{t}_{iW} \tag{17}$$

In this case, d_{iW} is the distance to wall W, with $\mathbf{n}_{iW}$ as the direction perpendicular to it, and $\mathbf{t}_{iW}$ as the direction tangential to it.

Most of the model parameters are taken from the well-established social force model (Helbing et al., 2000), which has been extensively validated in previous studies. The pedestrian mass, radius, and characteristic time are set to $m_i = 80$ kg, $r_i = 0.25$ m, and $\tau_i = 0.5$ s, respectively. The two constant parameters $A_i = 2\cdot10^3$ N and $B_i = 0.08$ m determine the strength and range of repulsive interaction between pedestrians. The body force coefficient $k = 1.2\cdot10^5$ kgs^{-2} and sliding friction coefficient $\kappa = 2.4\cdot10^5$ kgm^{-1}s^{-1} account for the obstructive effects from physical contact interactions. For both mainstream and moving (i.e., inverse and crossing) non-mainstream pedestrians entering the domain from the boundaries, their desired speeds are set to $v_i^0 = 1$ ms^{-1}, and desired directions $\mathbf{e}_i^0$ are uniformly sampled within an angular range $\theta_i \sim \mathcal{U}(-\theta^*, \theta^*)$ around the inward boundary normal $\mathbf{n}_b$. For static non-mainstream pedestrians, their desired speeds are set to $v_i^0 = 0$ ms^{-1}, and accordingly, desired directions are defined as a null vector $\mathbf{e}_i^0 = \mathbf{0}$.

3. Results

3.1. Identification of non-mainstream behaviors in real-world datasets

In pedestrian flows, deviations from the macroscopic mainstream are not random but are actively induced by individual decisions, and therefore, identifying non-mainstream behaviors is crucial for understanding the formation of congestion and predicting regions prone to conflicts. The classification based solely on entire trajectories provides temporal stability but obscures transient deviations, while frame-by-frame classification is sensitive to momentary fluctuations such as gait oscillations or brief avoidance maneuvers, resulting in unstable outcomes. To overcome these limitations, we adopt a segment-based approach within a spatial grid cell, which effectively balances temporal stability with local sensitivity and enables a robust identification of non-mainstream behaviors.

The fundamental question we address here is what drives the emergence of congestion in pedestrian flows. Intuitively, non-mainstream behaviors (e.g., inverse, crossing, and static) may disrupt the dominant flow and contribute to local crowding and crowd risk. To examine this hypothesis, Fig. 2(a) and 2(d) first visualize the identified mainstream and non-mainstream trajectory segments in the two real-world datasets. Mainstream behaviors (blue lines) form the dominant directional flow, whereas non-mainstream behaviors (red lines) appear as localized inverse, crossing, or lingering movements. Echoing these trajectory patterns, we next examine the corresponding spatial crowd states. Fig. 2(b) and 2(e) illustrate the spatial distributions of congestion level using the metric proposed by Feliciani and Nishinari (2018), while Fig. 2(c) and 2(f) further show crowd danger, which incorporates local density into the congestion measure and provides a complementary representation of spatial crowd risk. In both datasets, regions with pronounced non-mainstream activity frequently coincide with localized areas of elevated congestion and crowd danger, although the spatial patterns differ between the two contexts. These observations suggest a spatial association between non-mainstream behaviors and degraded local crowd conditions, motivating the subsequent quantitative analyses of their effects on pedestrian flow dynamics.

To further quantify the statistical signatures of these identified behaviors, Fig. 3 summarizes the spatio-temporal characteristics of mainstream and non-mainstream behaviors in both LargeView_Zoom_A and LargeView_Zoom_O. As shown in Fig. 3(a) and 3(d), mainstream pedestrians consistently constitute the majority of the crowd throughout the observation period. However, non-mainstream pedestrians maintain a non-negligible minority presence, coexisting dynamically with the dominant flow rather than appearing as isolated, momentary anomalies. The walking speed distributions in Fig. 3(b) and 3(e) demonstrate a clear kinematic separation between the two groups: mainstream behaviors generally maintain higher walking speeds, whereas non-mainstream behaviors are systematically slower, with their distributions skewed toward the lower end. This substantial speed reduction reveals the inherently constrained nature of pedestrian movement when navigating against, crossing, or lingering within a dense pedestrian flow. Furthermore, the direction-angle distributions in Fig. 3(c) and 3(f) provide robust validation of the identification algorithm. Mainstream trajectories cluster sharply around 90°, perfectly representing the unified macroscopic flow direction. In contrast, non-mainstream behaviors exhibit a broad, scattered angular distribution across the entire spectrum, which confirms their multidirectional and disruptive nature. Importantly, both cases exhibit highly consistent statistical patterns, indicating that the identified non-mainstream behaviors are robust across different periods in real-world environments.

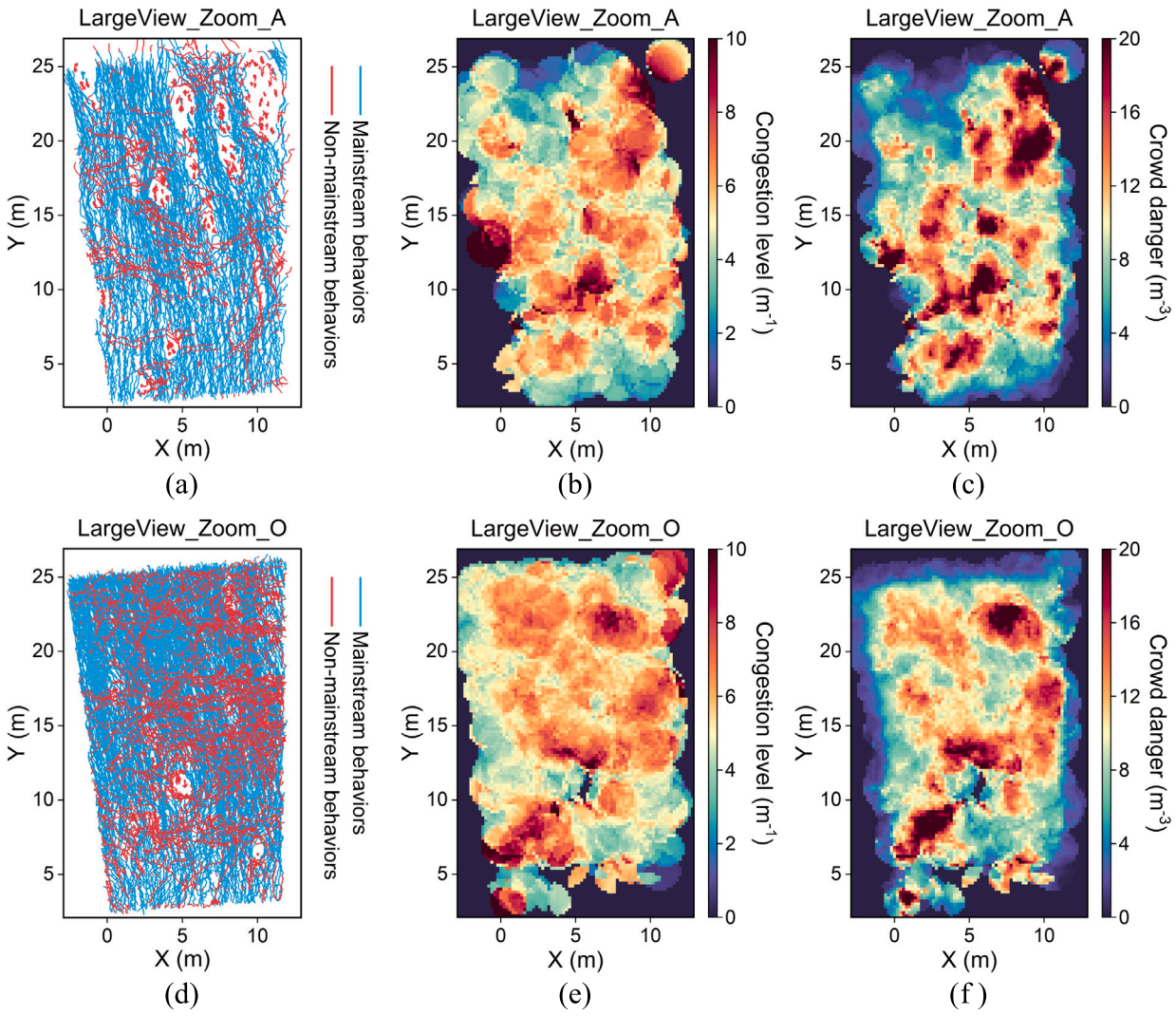


Fig. 2. Visualization of identified pedestrian behaviors and crowd states in real-world datasets LargeView_Zoom_A and LargeView_Zoom_O. (a, d) Identification results of mainstream and non-mainstream behaviors. (b, e) Spatial distributions of the congestion level. (c, f) Spatial distributions of the crowd danger.

3.2. Microscopic effects of non-mainstream behaviors on pedestrian flows

To understand the microscopic mechanisms through which non-mainstream behaviors disrupt pedestrian flows, we evaluate the local dynamic states surrounding different behavioral types. Specifically, three microscopic analytical metrics (i.e., local density, local speed variability, and local flow) are defined to quantify the spatial compression, kinetic fluctuation, and movement efficiency during local interactions, respectively. Afterwards, we also investigate how the non-mainstream ratio within a spatial cell affects the corresponding congestion level, thereby highlighting it as a potential driver beyond mere density effects.

Fig. 4 presents the statistical comparison of microscopic analytical metrics between mainstream and non-mainstream behaviors in real-world datasets. The group differences are assessed separately for each metric and dataset using two-sided Mann–Whitney U tests. As shown in Fig. 4(a) and 4(d), the local density around non-mainstream behaviors exhibits context-dependent differences ($P < 0.001$), being slightly higher in Fig. 4(a) but slightly lower in Fig. 4(d) compared to the mainstream group. This contrast indicates that local density may be composition-dependent, with stationary or lingering behaviors promoting accumulation, while inverse and crossing behaviors induce dispersion and lower instantaneous density. Next, Fig. 4(b) and 4(e) illustrate that non-mainstream behaviors are associated with a substantially higher local speed variability ($P < 0.001$). This elevated variability reflects the frequent stop-and-go motions, sharp steering, and avoidance maneuvers triggered by counter-flow or lateral conflicts, which inherently amplify local disorder. Finally, Fig. 4(c) and 4(f) demonstrate that the local flow around non-mainstream behaviors is markedly lower ($P < 0.001$). This significant reduction confirms that non-mainstream behaviors severely degrade microscopic movement efficiency, dissipating system momentum into social friction. Overall, these findings reveal a consistent mechanistic footprint: while the impact on local density depends on the specific behavioral composition, non-mainstream behaviors robustly increase dynamical instability and reduce capacity in pedestrian flows.

Further, we quantitatively analyze the relationship between cell-level non-mainstream ratio and congestion level. To control the potential influence of heterogeneous pedestrian flux, we adopt the cell segment frequency to quantify how frequently a grid cell is occupied or traversed by pedestrians. Fig. 5(a)–(b) display the probability density distributions of segment frequencies for the two contexts, with the dashed boundaries f_1 and f_2 selected at approximately the one-third and two-thirds ranks of the ordered non-missing frequencies. This rank-based criterion is intended to obtain low-, medium-, and high-frequency groups with approximately equal numbers of cells to

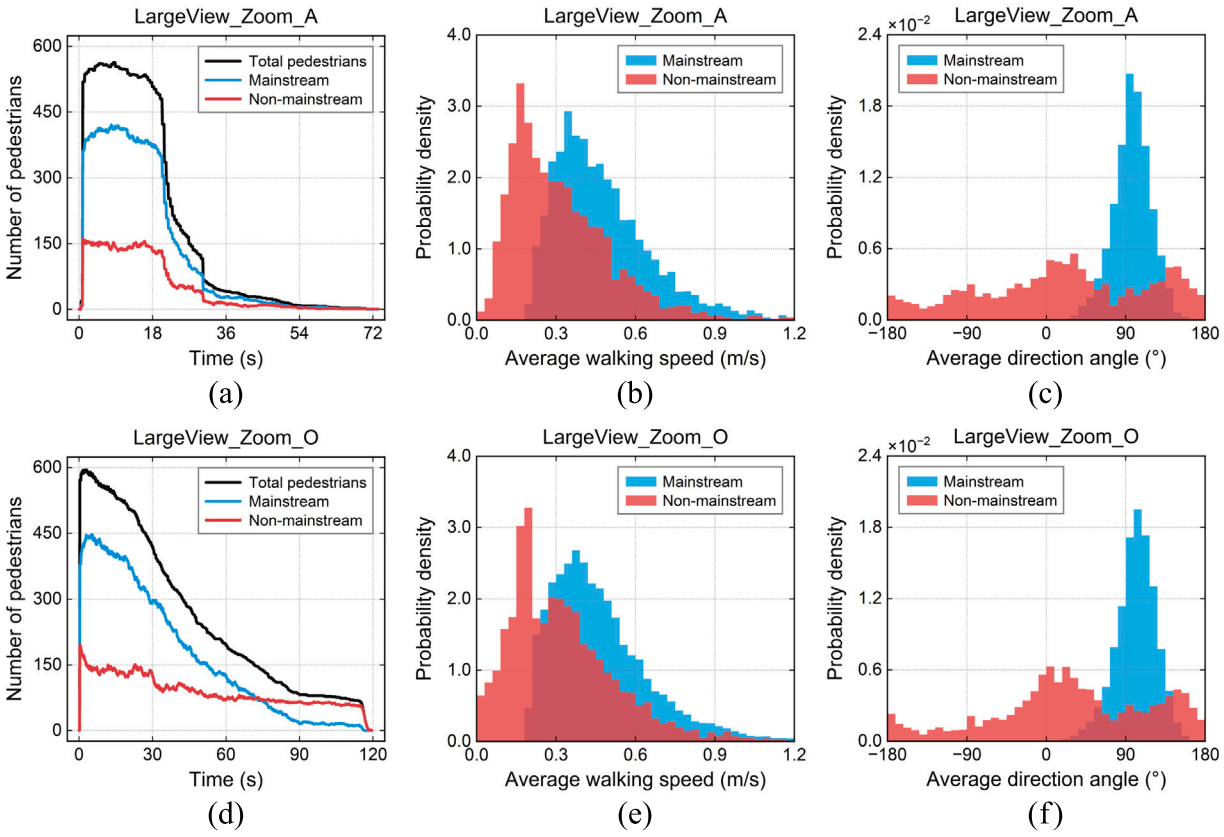


Fig. 3. Statistical analysis of mainstream and non-mainstream behaviors in real-world datasets LargeView_Zoom_A and LargeView_Zoom_O. (a, d) Numbers of pedestrians as a function of time. (b, e) Distributions of the average walking speed. (c, f) Distributions of the average direction angle.

facilitate subsequent comparisons. The resulting thresholds (f_1, f_2) are $(0.23\ \mathrm{s}^{-1}, 0.33\ \mathrm{s}^{-1})$ for LargeView_Zoom_A and $(0.24\ \mathrm{s}^{-1}, 0.40\ \mathrm{s}^{-1})$ for LargeView_Zoom_O, respectively. Based on this division, Fig. 5(c)–(d) show the relationship between cell non-mainstream ratio and cell congestion level under different frequency regimes. A clear positive correlation is consistently observed across both contexts (black lines), indicating that higher proportions of non-mainstream behaviors systematically correspond to elevated congestion levels. Notably, this trend persists across all frequency categories, which implies that the impact of non-mainstream behaviors on congestion is robust with respect to variations in local pedestrian flux. To further assess threshold sensitivity, we independently vary f_1 and f_2 by $\pm 10\%$ in 5% increments. Across all 25 combinations per dataset, the association remains positive and statistically significant in all three frequency groups after Benjamini–Hochberg correction. This reveals that congestion is associated not only with local pedestrian flux but also with behavioral composition, with higher non-mainstream ratios consistently corresponding to higher congestion levels.

3.3. Macroscopic effects of density and non-mainstream ratio in simulations

Although the preceding empirical analysis reveals the statistical signatures and localized disruptions of non-mainstream behaviors, real-world trajectory data are inevitably constrained by specific environmental layouts and pedestrian flows. To systematically evaluate the macroscopic impacts of endogenous perturbations under controlled conditions, we develop an adapted social force model to conduct numerical simulations. In this model, the crowd density is controlled via the spawn probability (P_s), which determines the likelihood of generating a pedestrian at each time step, while the non-mainstream ratio is controlled via the non-mainstream probability (P_n), which specifies the likelihood that a newly generated pedestrian exhibits non-mainstream behavior, with equal shares assigned to left-side crossing, right-side crossing, inverse, and static pedestrians. An auxiliary comparison with the two empirical datasets confirms that, after limited calibration, the model reproduces the main macroscopic flow characteristics within the observed range. Building on this consistency check, we independently vary these two probabilities to examine how they reshape collective flow dynamics under controlled conditions.

Fig. 6 illustrates the schematic diagram of the simulation scenario and presents typical snapshots under different parameter combinations. As depicted in Fig. 6(a), the simulation scenario is designed to represent the principal directional configurations observed in the empirical pedestrian flows, where mainstream pedestrians consistently move from the bottom to the top, while non-mainstream pedestrians are spawned from other boundaries or within the corridor itself, including inverse, crossing, and static behaviors. Fig. 6(b) visualizes a simulated snapshot under low crowd density and non-mainstream ratio ($P_s = 0.3$, $P_n = 0.1$). Under this sparse condition, the system maintains a relatively ordered state. Mainstream pedestrians can easily identify spatial gaps to navigate around the minority of non-mainstream individuals, resulting in smooth movement trajectories. Conversely, when both crowd density and non-mainstream ratio are significantly increased ($P_s = 0.8$, $P_n = 0.4$) in Fig. 6(c), the collective flow structure breaks down, and the snapshot reveals severe spatial mixing, widespread trajectory entanglement, and the formation of localized blockages. These examples demonstrate that our model can flexibly capture a wide range of flow regimes through parameter control, providing an effective framework for analyzing the influences of density level and non-mainstream ratio on pedestrian flows.

To systematically quantify the macroscopic effects observed in the snapshots, we scan these two parameters as the spawn probability $P_s = 0.1, 0.2, \ldots, 1.0$ and the non-mainstream probability $P_n =$

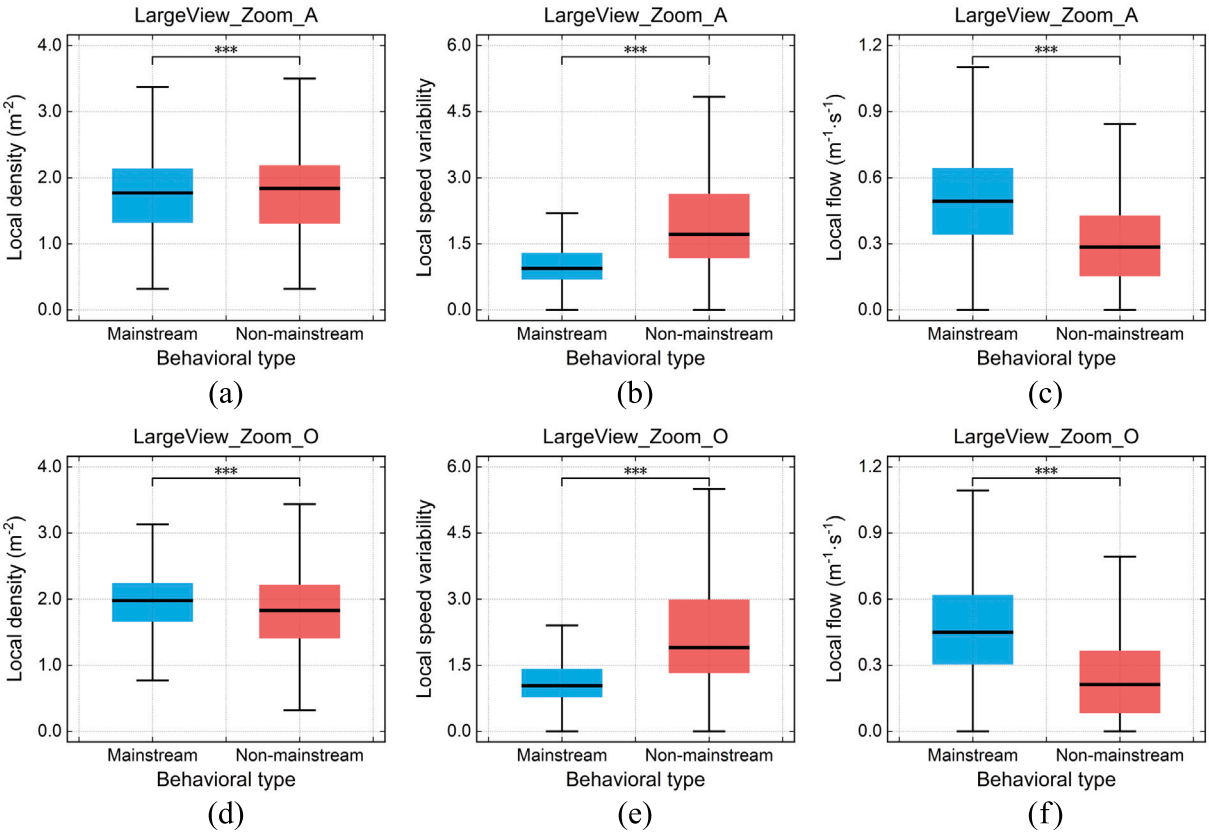


Fig. 4. Microscopic comparison of mainstream and non-mainstream behaviors in real-world datasets LargeView_Zoom_A and LargeView_Zoom_O. (a, d) Boxplots of the local density. (b, e) Boxplots of the local speed variability. (c, f) Boxplots of the local flow. *** $P < 0.001$.

$0.05, 0.1, \ldots, 0.5$. Fig. 7 shows the resulting heatmaps for different analytical metrics, where the kinematic responses of the crowd are first evaluated in Fig. 7(a)–7(c) . From a horizontal perspective, increasing the spawn probability consistently leads to higher density, lower speed, and larger flow, reflecting the well-established fundamental diagram relationships in pedestrian dynamics. From a vertical perspective, increasing the non-mainstream probability also results in higher density and lower speed, but, in contrast to the density-driven trend, leads to a reduction in flow, and these trends become more pronounced at higher crowd densities. This indicates that non-mainstream behaviors introduce additional friction that disrupts efficient movement and suppresses system throughput. To evaluate macroscopic crowd safety, the order parameter Φ measures directional alignment based on the normalized velocities of moving pedestrians (Vicsek et al., 1995). The congestion level Cl quantifies local velocity-field curl variation relative to the mean walking speed (Feliciani and Nishinari, 2018), while crowd danger Cd further incorporates local density ρ to represent density-weighted flow disorganization. The safety metrics in Fig. 7(d)–7(f) further underscore the destabilizing mechanisms of non-mainstream behaviors. Notably, the average order parameter shows clear horizontal stratification, it remains high at lower P_n but drops sharply with increasing P_n, largely independent of density, indicating that the loss of order is primarily driven by behavioral diversity rather than crowding. Moreover, the congestion level and crowd danger both peak in the top-right quadrant (high P_s and P_n). This represents a synergistic deterioration: while density provides the physical basis for compression, the injection of non-mainstream behaviors acts as the critical trigger that converts a dense but ordered flow into a highly congested and dangerous state.

3.4. Structural control strategies for regulating pedestrian flows

Given the potential destabilizing effects of crowd density and non-mainstream ratio, it is practically imperative to explore mitigation measures to enhance flow resilience. Motivated by previous experimental studies showing that strategically placed obstacles near exits can paradoxically improve pedestrian flows (Shiwakoti et al., 2019), we introduce physical structural interventions into our simulation scenario as control strategies. These interventions are designed to regulate mixed crowd dynamics and alleviate chaotic interactions between mainstream and non-mainstream pedestrians, thereby improving overall flow efficiency and system resilience.

To untangle mixed flows, two structural control strategies, CS-Diamond and CS-Circle, are designed to place diamond-shaped and circular-shaped zones at the center of the scenario, respectively. The diamond and circle have a half-diagonal and radius of 2.5 m, and serve as static lingering areas where stationary pedestrians are concentrated, resembling small visitor zones in reality. The black boundaries surrounding these areas act as walls separating the intervention zones from the dynamic pedestrian stream. Pedestrians interact with these boundaries through the pedestrian–wall force defined in Eq. (17), whose repulsive and contact components prevent penetration and redirect movement around the prescribed geometries. Fig. 8 visualizes the simulation snapshots of different control strategies under moderate crowd density and non-mainstream ratio ($P_s = 0.6, P_n = 0.3$). For the Baseline scenario in Fig. 8(a), without structural intervention, mainstream and non-mainstream trajectories are heavily entangled due to unconstrained physical conflicts, producing widespread spatial disorder. In contrast, Fig. 8(b)–(c) demonstrate that CS-Diamond and CS-Circle partition the scenario into distinct flow regions. The interventions force pedestrians to navigate around the predefined geometries, naturally channeling opposing flows into distinct localized lanes, which reduces

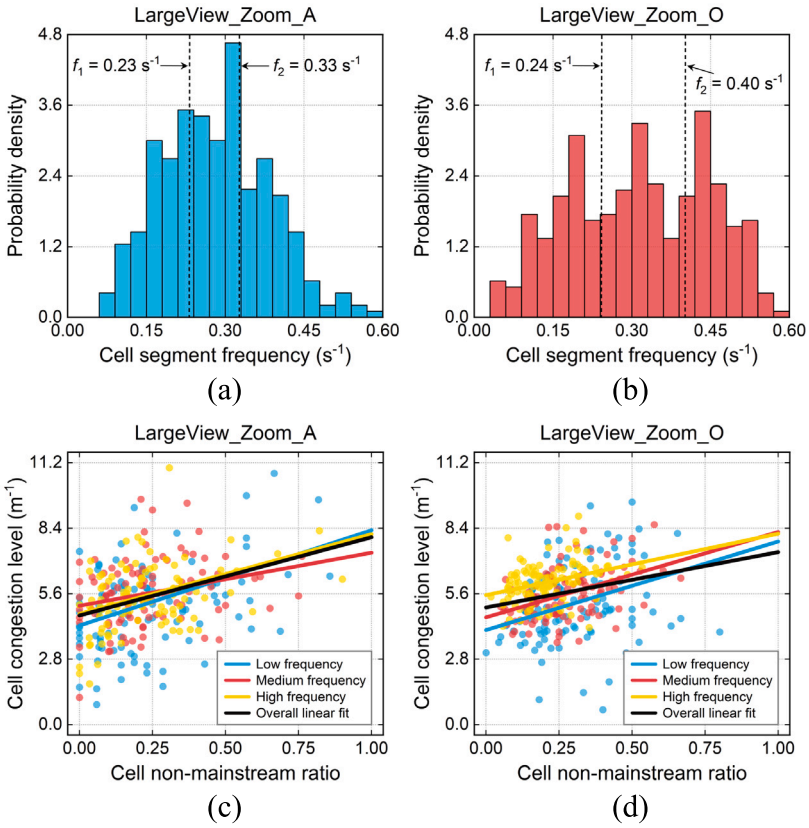


Fig. 5. Quantitative analysis of the relationship between cell-level non-mainstream ratio and congestion level in real-world datasets LargeView_Zoom_A and LargeView_Zoom_O. (a–b) Distributions of the cell segment frequency. (c–d) Cell congestion level as a function of the cell non-mainstream ratio.

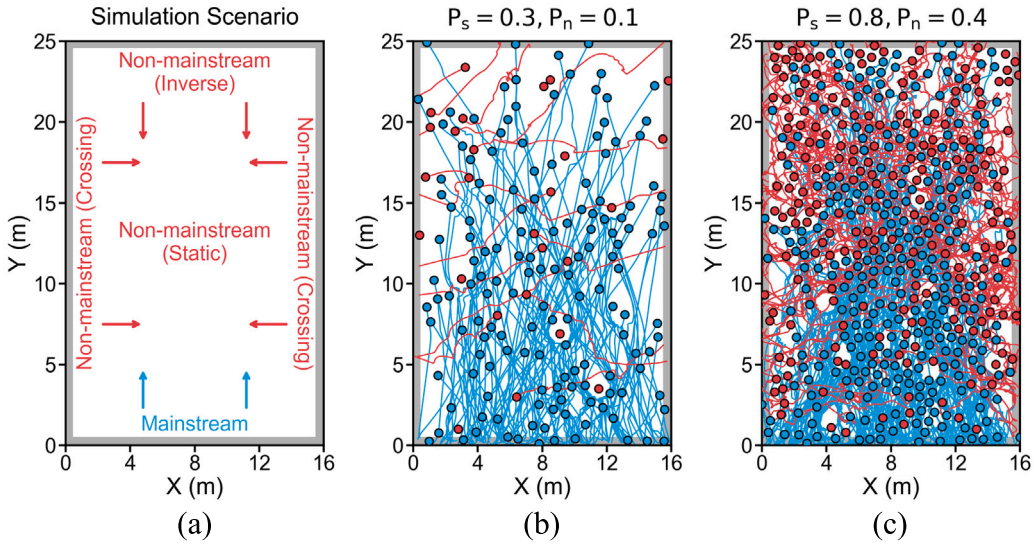


Fig. 6. Illustration of simulation scenario and snapshots. (a) Schematic diagram of the simulation scenario. (b) Simulation snapshot under low crowd density and non-mainstream ratio (P_s = 0.3, P_n = 0.1). (c) Simulation snapshot under high crowd density and non-mainstream ratio (P_s = 0.8, P_n = 0.4). Mainstream and non-mainstream pedestrians and their trajectories are represented by blue and red circles and lines, respectively.

head-on collisions and trajectory entanglement. This visual comparison, while preliminary, provides evidence that appropriately designed structural controls can effectively restore a degree of collective order.

To perform a quantitative assessment of the proposed control strategies, Figs. 9 and 10 present the macroscopic metrics as functions of crowd density and non-mainstream ratio, respectively, where each curve is obtained by averaging over the other variable. Despite differing roles of the two parameters, both figures exhibit broadly consistent intervention effects. As either crowd density or non-mainstream ratio increases, compared to the Baseline, both CS-Diamond and CS-Circle lead to a slight increase in average density, a significant enhancement in average speed and flow, a moderate improvement in collective

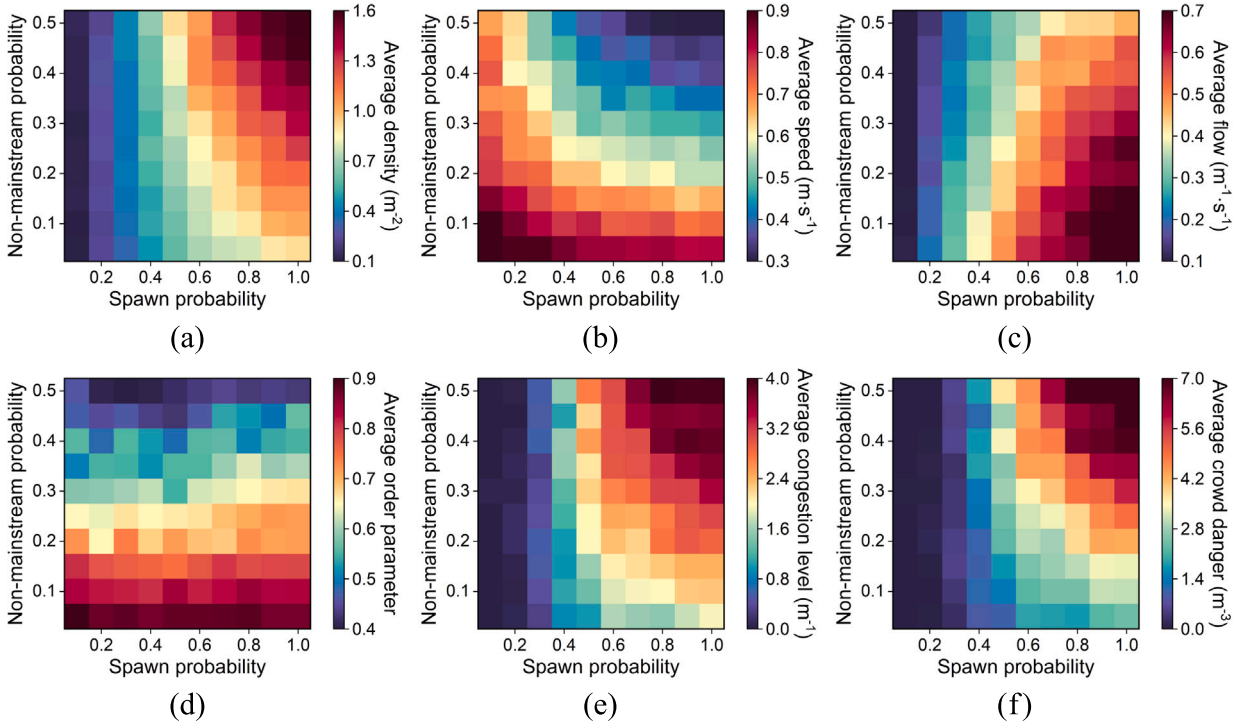


Fig. 7. Macroscopic effects of crowd density and non-mainstream ratio on pedestrian flows in simulations. (a–c) Kinematic metrics: average density, average speed, and average flow. (d–f) Safety metrics: average order parameter, average congestion level, and average crowd danger.

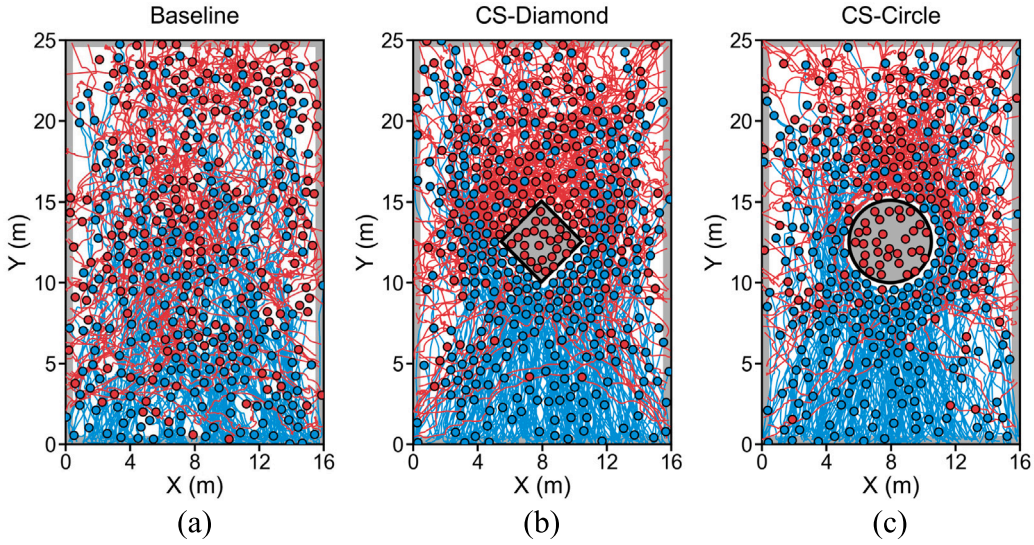


Fig. 8. Simulation snapshots of different control strategies under moderate crowd density and non-mainstream ratio ($P_s = 0.6, P_n = 0.3$). (a) Baseline (no control strategy). (b) CS-Diamond. (c) CS-Circle. Mainstream and non-mainstream pedestrians and their trajectories are represented by blue and red circles and lines, respectively.

order, and an apparent reduction in average congestion level and crowd danger. These improvements may reflect two complementary mechanisms: spatially separating stationary pedestrians from moving streams and geometrically guiding opposing and crossing movements around designated lingering areas. This reorganization is expected to reduce conflicting encounters and associated braking and avoidance responses without altering underlying interaction laws. The improvements in movement efficiency are consistent with experiments showing that roundabouts can increase walking speeds at intersections despite reduced space and longer paths (Sun et al., 2018). Notably, the two strategies exhibit comparable impacts on kinematic metrics, while CS-Circle outperforms CS-Diamond in safety metrics. A plausible explanation is that the circular boundary allows smoother directional adjustments, whereas the diamond's sharp corners may induce local accumulation and intermittent bottlenecks. These findings suggest that appropriately designed structural interventions can improve movement efficiency and potentially alleviate congestion and risk, although their

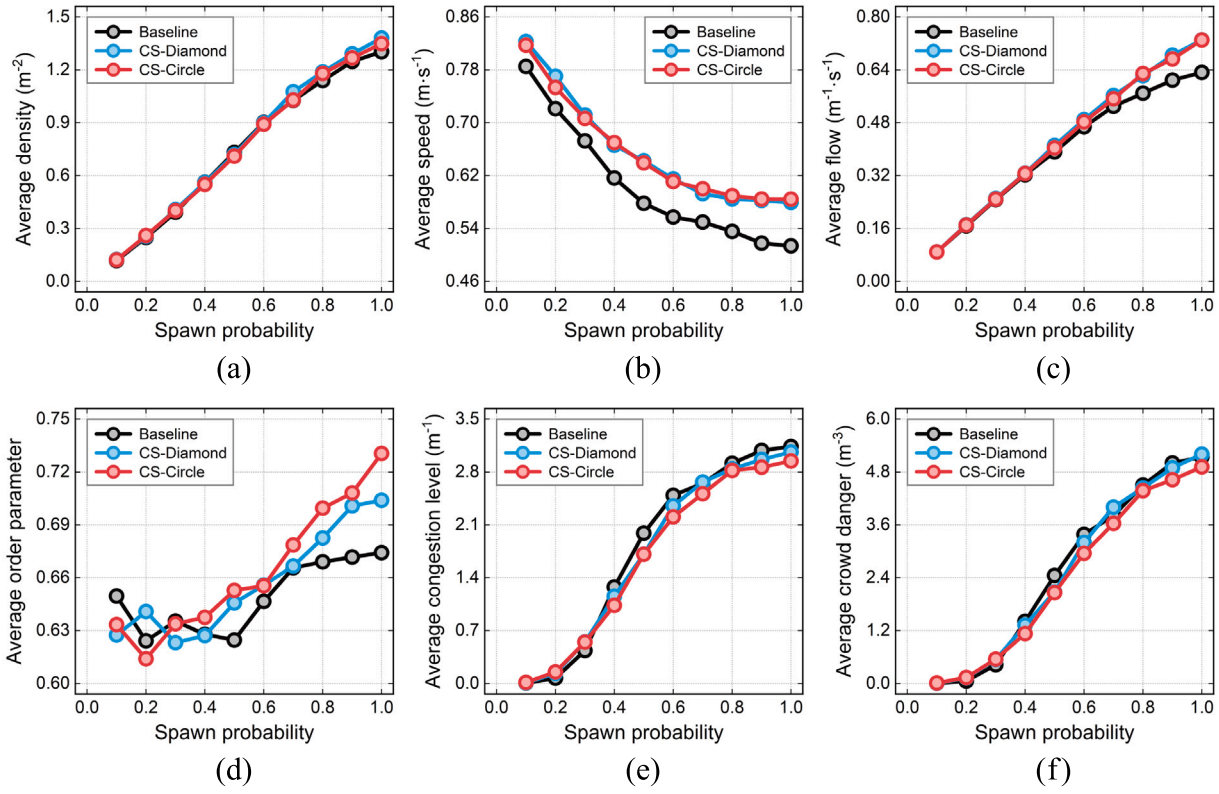


Fig. 9. Macroscopic metrics as a function of crowd density under different structural control strategies. (a–c) Kinematic metrics: average density, average speed, and average flow. (d–f) Safety metrics: average order parameter, average congestion level, and average crowd danger.

effectiveness depends on geometric and crowd conditions (Shiwakoti et al., 2019) and requires further experimental validation.

4. Conclusions

In this work, we establish a comprehensive framework to investigate how non-mainstream behaviors influence pedestrian flow dynamics, spanning from empirical data analysis to computational simulations. The main conclusions can be summarized as follows: (1) The proposed segment-based spatial cell method effectively identifies non-mainstream behaviors, and subsequent statistical analyses confirm that the identified behaviors are robust and consistent across different contexts. (2) At the microscopic level, non-mainstream behaviors systematically elevate local speed variability and decrease local flow, with a consistent positive correlation observed between their ratio and localized congestion within each grid cell. (3) At the macroscopic level, simulation results show that higher crowd density raises compression and flow demand, while greater non-mainstream ratio introduces additional friction that slows movement and destabilizes order, and these factors synergistically amplify both congestion and crowd risk. (4) Structural control strategies, such as geometric interventions, reorganize mixed flows, mitigate conflicts, and improve both efficiency and safety to some extent, providing simulation-based insights into the regulation of complex pedestrian flows.

This work also provides valuable practical implications for monitoring, managing, and optimizing dense pedestrian flows in real-world environments. First, the segment-based identification method combined with localized microscopic metrics offers an operational tool for real-time detection of behavioral deviations, allowing operators in subway stations, stadiums, or urban squares to identify regions prone to non-mainstream movements and issue early warnings before local congestion occurs (Liu et al., 2025). Second, the observed positive correlation between the ratio of non-mainstream behaviors and congestion level informs proactive crowd control strategies: targeted interventions such as temporary route planning, staff deployment, or guidance signage can reduce localized breakdowns and improve efficiency during peak hours or large events (Zhang et al., 2016, 2021; Wang et al., 2025). Third, the evaluation of structural control strategies enables actionable guidance for facility planning and event management: appropriately placing geometric obstacles, temporary visitor zones, or directional cues may help channel pedestrian movement, minimize conflicts, and enhance throughput and resilience in mixed-flow scenarios (Yu et al., 2025; Wang et al., 2019; Lin et al., 2020). Collectively, these insights bridge theoretical understanding with practical engineering, supporting the design of safer infrastructure, optimized crowd flows, and evidence-based operational protocols.

Furthermore, several limitations of this study should be emphasized to guide future research. First, the identification method relies on predefined thresholds for speed and angular deviation within grid cells, which may not capture highly complex or context-dependent maneuvers in extremely turbulent crowd conditions (Helbing et al., 2007; Gu et al., 2025). Second, our adapted social force model does not explicitly model cognitive or anticipatory interactions (Moussaïd et al., 2011; Yi et al., 2023), which may reduce the smoothness and coordination of avoidance behaviors in dense or complex flows compared with real pedestrian movement. Third, the structural control strategies evaluated in this study are limited to specific static geometries (diamond and circle) located at the center of the corridor, but the size, number, and layout of such interventions under varying architectural environments remain to be fully explored (Shiwakoti et al., 2019). Future work could extend the current framework by introducing adaptive intention-recognition algorithms, integrating collision-prediction rules into the force-based models, and optimizing structural interventions using machine learning. Despite these limitations, this work still deepens our understanding of behavior-induced congestion in pedestrian flows and offers constructive strategies for real-world crowd regulation.

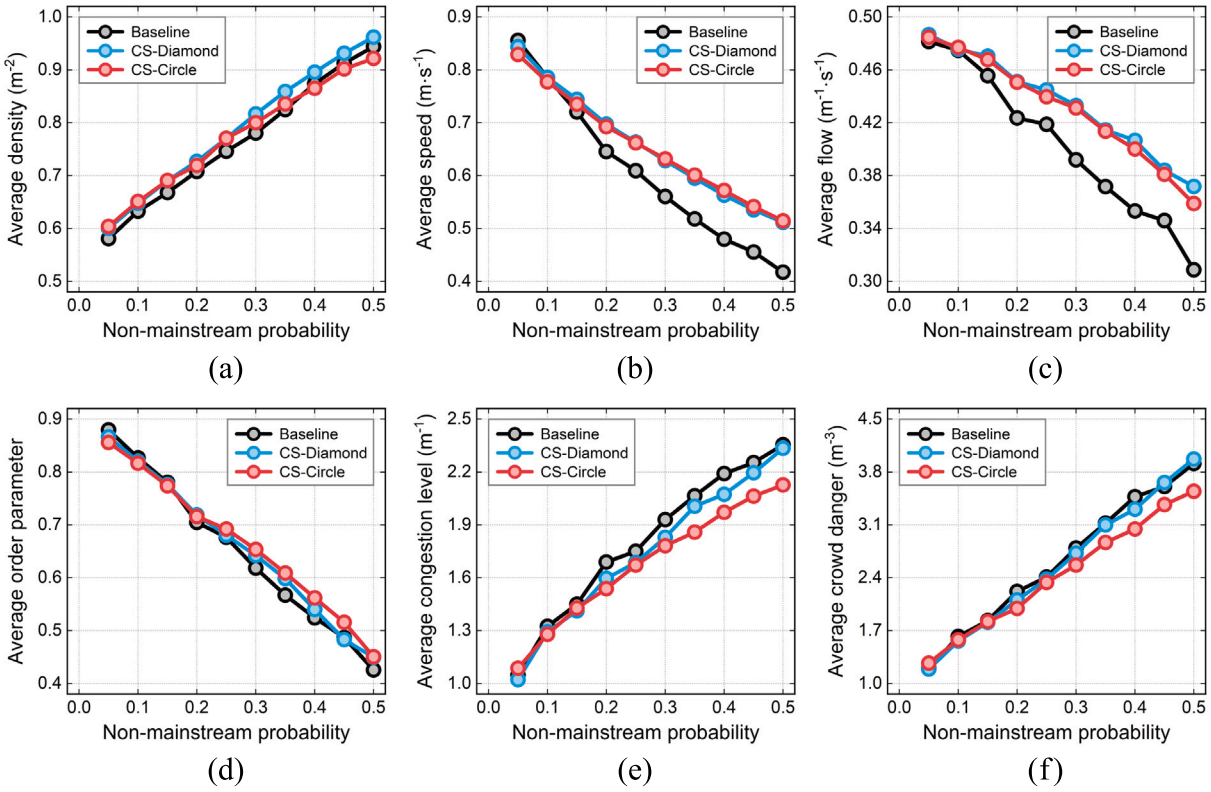


Fig. 10. Macroscopic metrics as a function of non-mainstream ratio under different structural control strategies. (a–c) Kinematic metrics: average density, average speed, and average flow. (d–f) Safety metrics: average order parameter, average congestion level, and average crowd danger.

CRediT authorship contribution statement

Wenhan Wu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Wenfeng Yi:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The empirical trajectory data are described in the data descriptor (Dufour et al., 2025).

References

Chen, S., Fu, L., Fang, J., Yang, P., 2019. The effect of obstacle layouts on pedestrian flow in corridors: An experimental study. Phys. A (ISSN: 0378-4371) 534, 122333. http://dx.doi.org/10.1016/j.physa.2019.122333.

Ding, N., Liu, T., Zhang, Z., Xu, Y., 2025. The impact of obstacles on the pedestrian flow of middle and high school students: A controlled experimental investigation. Phys. A (ISSN: 0378-4371) 675, 130833. http://dx.doi.org/10.1016/j.physa.2025.130833.

Dufour, O., Dang, H.-T., Cordes, J., Korbmacher, R., Chraibi, M., Gaudou, B., Nicolas, A., Tordeux, A., 2025. Dense crowd dynamics and pedestrian trajectories: A multiscale field dataset from the festival of lights in lyon. Sci. Data (ISSN: 2052-4463) 12 (1), http://dx.doi.org/10.1038/s41597-025-04732-3.

Feliciani, C., Nishinari, K., 2016. Empirical analysis of the lane formation process in bidirectional pedestrian flow. Phys. Rev. E (ISSN: 2470-0053) 94 (3), 032304. http://dx.doi.org/10.1103/physreve.94.032304.

Feliciani, C., Nishinari, K., 2018. Measurement of congestion and intrinsic risk in pedestrian crowds. Transp. Res. Part C: Emerg. Technol. (ISSN: 0968-090X) 91, 124–155. http://dx.doi.org/10.1016/j.trc.2018.03.027.

Fu, T., Xie, S., Li, R., Wang, J., Zhang, L., Sobhani, A., Fang, S., 2023. Investigating pedestrian crossing patterns at crossing locations based on trajectory data collected by UAV. In: Lee, S. (Ed.), J. Adv. Transp. (ISSN: 2042-3195) 2023 (1), http://dx.doi.org/10.1155/2023/5529970.

Garcimartín, A., Maza, D., Pastor, J.M., Parisi, D.R., Martín-Gómez, C., Zuriguel, I., 2018. Redefining the role of obstacles in pedestrian evacuation. New J. Phys. (ISSN: 1367-2630) 20 (12), 123025. http://dx.doi.org/10.1088/1367-2630/aaf4ca.

Gu, F., Guiselin, B., Bain, N., Zuriguel, I., Bartolo, D., 2025. Emergence of collective oscillations in massive human crowds. Nature (ISSN: 1476-4687) 638 (8049), 112–119. http://dx.doi.org/10.1038/s41586-024-08514-6.

Helbing, D., Farkas, I., Vicsek, T., 2000. Simulating dynamical features of escape panic. Nature (ISSN: 1476-4687) 407 (6803), 487–490. http://dx.doi.org/10.1038/35035023.

Helbing, D., Johansson, A., Al-Abideen, H.Z., 2007. Dynamics of crowd disasters: An empirical study. Phys. Rev. E (ISSN: 1550-2376) 75 (4), 046109. http://dx.doi.org/10.1103/physreve.75.046109.

Hou, J., Zhang, E., Long, Y., 2023. Measuring pedestrian flows in public spaces: Inferring walking for transport and recreation using Wi-Fi probes. Build. Environ. (ISSN: 0360-1323) 230, 109999. http://dx.doi.org/10.1016/j.buildenv.2023.109999.

Hu, Y., Bi, Y., Li, H., Gao, W., Zhang, J., Song, W., 2023. An empirical study on the effect of an obstacle on the inflow process. Phys. A (ISSN: 0378-4371) 621, 128788. http://dx.doi.org/10.1016/j.physa.2023.128788.

Hu, Y., Li, C., Ren, X., Li, H., 2025. The impact of a pedestrian obstacle at different times at the bottleneck on evacuation. Phys. A (ISSN: 0378-4371) 660, 130325. http://dx.doi.org/10.1016/j.physa.2024.130325.

Jia, X., Murakami, H., Feliciani, C., Yanagisawa, D., Nishinari, K., 2021. Pedestrian lane formation and its influence on egress efficiency in the presence of an obstacle. Saf. Sci. (ISSN: 0925-7535) 144, 105455. http://dx.doi.org/10.1016/j.ssci.2021.105455.

Jia, X., Yue, H., Tian, X., Yin, H., 2017. Simulation of pedestrian flow with evading and surpassing behavior in a walking passageway. Simulation (ISSN: 1741-3133) 93 (12), 1013–1035. http://dx.doi.org/10.1177/0037549717734633.

Jiang, L., Li, J., Shen, C., Yang, S., Han, Z., 2014. Obstacle optimization for panic flow - Reducing the tangential momentum increases the escape speed. In: Gao, Z.-K. (Ed.), PLoS One (ISSN: 1932-6203) 9 (12), e115463. http://dx.doi.org/10.1371/journal.pone.0115463.

Li, B., Qu, Y., Xiao, Y., Wu, J., Gao, Z., 2025. A geometric cognition pedestrian dynamics model considering heterogeneous collision avoidance behaviors. Transp. Res. Part B: Methodol. (ISSN: 0191-2615) 200, 103309. http://dx.doi.org/10.1016/j.trb.2025.103309.

Lin, J., Zhu, R., Li, N., Becerik-Gerber, B., 2020. Do people follow the crowd in building emergency evacuation? A cross-cultural immersive virtual reality-based study. Adv. Eng. Informatics (ISSN: 1474-0346) 43, 101040. http://dx.doi.org/10.1016/j.aei.2020.101040.

Liu, Y., Ma, X., Tao, Y., Dong, L., Ding, X., Qiu, X., 2024. Numerical investigation on the impact of obstacles on phase transition in pedestrian counter-flow. Phys. A (ISSN: 0378-4371) 635, 129499. http://dx.doi.org/10.1016/j.physa.2024.129499.

Liu, S., Zhang, L., Ge, J., Li, W., Long, Y., 2025. Framework for timely perception of spatiotemporal crowd congestion risk in public spaces based on video pedestrian tracking and geographic mapping. GIScience & Remote. Sens. (ISSN: 1943-7226) 62 (1), http://dx.doi.org/10.1080/15481603.2025.2480416.

Ma, J., Song, W.G., Lo, S.M., Fang, Z.M., 2013. New insights into turbulent pedestrian movement pattern in crowd-quakes. J. Stat. Mech. Theory Exp. (ISSN: 1742-5468) 2013 (02), P02028. http://dx.doi.org/10.1088/1742-5468/2013/02/p02028.

Masumura, A., Sadohara, S., 2023. Proposal of a classification method for public space usage based on pedestrian trajectory data. Jpn. Archit. Rev. (ISSN: 2475-8876) 6 (1), http://dx.doi.org/10.1002/2475-8876.12359.

McArdle, G., Demšar, U., van der Spek, S., McLoone, S., 2014. Classifying pedestrian movement behaviour from GPS trajectories using visualization and clustering. Ann. GIS (ISSN: 1947-5691) 20 (2), 85–98. http://dx.doi.org/10.1080/19475683.2014.904560.

Moussaïd, M., Guillot, E.G., Moreau, M., Fehrenbach, J., Chabiron, O., Lemercier, S., Pettré, J., Appert-Rolland, C., Degond, P., Theraulaz, G., 2012. Traffic instabilities in self-organized pedestrian crowds. In: Ben-Jacob, E. (Ed.), PLoS Comput. Biol. (ISSN: 1553-7358) 8 (3), e1002442. http://dx.doi.org/10.1371/journal.pcbi.1002442.

Moussaïd, M., Helbing, D., Theraulaz, G., 2011. How simple rules determine pedestrian behavior and crowd disasters. Proc. Natl. Acad. Sci. (ISSN: 1091-6490) 108 (17), 6884–6888. http://dx.doi.org/10.1073/pnas.1016507108.

Qiu, H., Fang, Z., Lee, E.W.M., 2025. Simulation study on the effect of obstacles upstream of the building exit on evacuation efficiency. Phys. A (ISSN: 0378-4371) 666, 130547. http://dx.doi.org/10.1016/j.physa.2025.130547.

Sharma, D., Bhondekar, A.P., Shukla, A.K., Ghanshyam, C., 2016. A review on technological advancements in crowd management. J. Ambient. Intell. Humaniz. Comput. (ISSN: 1868-5145) 9 (3), 485–495. http://dx.doi.org/10.1007/s12652-016-0432-x.

Sharma, A., McCloskey, B., Hui, D.S., Rambia, A., Zumla, A., Traore, T., Shafi, S., El-Kafrawy, S.A., Azhar, E.I., Zumla, A., Rodriguez-Morales, A.J., 2023. Global mass gathering events and deaths due to crowd surge, stampedes, crush and physical injuries – Lessons from the Seoul Halloween and other disasters. Travel. Med. Infect. Dis. (ISSN: 1477-8939) 52, 102524. http://dx.doi.org/10.1016/j.tmaid.2022.102524.

Shi, X., Ye, Z., Shiwakoti, N., Tang, D., Lin, J., 2019. Examining effect of architectural adjustment on pedestrian crowd flow at bottleneck. Phys. A (ISSN: 0378-4371) 522, 350–364. http://dx.doi.org/10.1016/j.physa.2019.01.086.

Shiwakoti, N., Shi, X., Ye, Z., 2019. A review on the performance of an obstacle near an exit on pedestrian crowd evacuation. Saf. Sci. (ISSN: 0925-7535) 113, 54–67. http://dx.doi.org/10.1016/j.ssci.2018.11.016.

Sticco, I., Frank, G., Dorso, C., 2022. Improving competitive evacuations with a vestibule structure designed from panel-like obstacles in the framework of the Social Force Model. Saf. Sci. (ISSN: 0925-7535) 146, 105544. http://dx.doi.org/10.1016/j.ssci.2021.105544.

Sun, L., Hao, S., Gong, Q., Qiu, S., Chen, Y., 2018. Pedestrian roundabout improvement strategy in subway stations. Proc. Inst. Civ. Eng. - Transp. (ISSN: 1751-7710) 171 (1), 20–29. http://dx.doi.org/10.1680/jtran.16.00073.

Vicsek, T., Czirók, A., Ben-Jacob, E., Cohen, I., Shochet, O., 1995. Novel type of phase transition in a system of self-driven particles. Phys. Rev. Lett. (ISSN: 1079-7114) 75 (6), 1226–1229. http://dx.doi.org/10.1103/physrevlett.75.1226.

Wang, J., Jin, B., Li, J., Chen, F., Wang, Z., Sun, J., 2019. Method for guiding crowd evacuation at exit: The buffer zone. Saf. Sci. (ISSN: 0925-7535) 118, 88–95. http://dx.doi.org/10.1016/j.ssci.2019.05.014.

Wang, J., Shen, L., Wang, H., Tong, Y., 2025. Investigating the impact of guide placement and signage type on pedestrian evacuation behavior using Virtual Reality experiments. J. Stat. Mech. Theory Exp. (ISSN: 1742-5468) 2025 (5), 053404. http://dx.doi.org/10.1088/1742-5468/add515.

Wu, W., Chen, M., Li, J., Liu, B., Zheng, X., 2022a. An extended social force model via pedestrian heterogeneity affecting the self-driven force. IEEE Trans. Intell. Transp. Syst. (ISSN: 1558-0016) 23 (7), 7974–7986. http://dx.doi.org/10.1109/tits.2021.3074914.

Wu, W., Li, J., Yi, W., Zheng, X., 2022b. Modeling crowd evacuation via behavioral heterogeneity-based social force model. IEEE Trans. Intell. Transp. Syst. (ISSN: 1558-0016) 23 (9), 15476–15486. http://dx.doi.org/10.1109/tits.2022.3140823.

Wu, X., Nie, L., Xu, M., Zhao, L., 2019. Distribution planning problem for a high-speed rail catering service considering time-varying demands and pedestrian congestion: A lot-sizing-based model and decomposition algorithm. Transp. Res. Part E: Logist. Transp. Rev. (ISSN: 1366-5545) 123, 61–89. http://dx.doi.org/10.1016/j.tre.2019.01.003.

Wu, W., Yi, W., 2026a. A methodological comparison of interaction neighborhoods in the social force model of panic evacuation. Reliab. Eng. Syst. Saf. (ISSN: 0951-8320) 274, 112457. http://dx.doi.org/10.1016/j.ress.2026.112457.

Wu, W., Yi, W., 2026b. Modeling the dynamical process of behavioral contagion in human crowds during evacuation. Reliab. Eng. Syst. Saf. (ISSN: 0951-8320) 266, 111649. http://dx.doi.org/10.1016/j.ress.2025.111649.

Wu, W., Yi, W., Wang, E., Wang, X., Zheng, X., 2025. How social attributes affect the movement process of subgroups when facing a static obstacle. IEEE Trans. Comput. Soc. Syst. (ISSN: 2373-7476) 12 (2), 658–670. http://dx.doi.org/10.1109/tcss.2024.3493954.

Wu, W., Zhang, K., Yi, W., 2026. Large-scale trajectory evidence reveals pedestrian speed regimes and behavioral thresholds in railway stations. Unpublished manuscript.

Yang, J., Zang, X., Chen, W., Luo, Q., Wang, R., Liu, Y., 2024. Improved social force model based on pedestrian collision avoidance behavior in counterflow. Phys. A (ISSN: 0378-4371) 642, 129762. http://dx.doi.org/10.1016/j.physa.2024.129762.

Yi, W., Wu, W., 2026. Control strategies for order–disorder phase transition in crowd evacuation. Reliab. Eng. Syst. Saf. (ISSN: 0951-8320) 266, 111688. http://dx.doi.org/10.1016/j.ress.2025.111688.

Yi, W., Wu, W., Wang, X., Wang, E., Zheng, X., 2024. Order–disorder phase transitions in front of the exit during human crowd evacuations. Transp. Res. Part C: Emerg. Technol. (ISSN: 0968-090X) 163, 104649. http://dx.doi.org/10.1016/j.trc.2024.104649.

Yi, W., Wu, W., Wang, X., Zheng, X., 2023. Modeling the mutual anticipation in human crowds with attention distractions. IEEE Trans. Intell. Transp. Syst. (ISSN: 1558-0016) 24 (9), 10108–10117. http://dx.doi.org/10.1109/tits.2023.3268315.

Yu, H., Jiang, N., Li, M., Jia, X., Shi, J., Wai Ming Lee, E., Yang, L., 2025. Regulating multi-directional passenger flow: Impact of obstacle position and flow level on pedestrian merging process. Tunn. Undergr. Space Technol. (ISSN: 0886-7798) 157, 106336. http://dx.doi.org/10.1016/j.tust.2024.106336.

Zeng, G., Schadschneider, A., Zhang, J., Wei, S., Song, W., Ba, R., 2019. Experimental study on the effect of background music on pedestrian movement at high density. Phys. Lett. A (ISSN: 0375-9601) 383 (10), 1011–1018. http://dx.doi.org/10.1016/j.physleta.2018.12.019.

Zhang, Z., Jia, L., Qin, Y., 2021. Location-allocation model for the design of guidance signage systems for pedestrian wayfinding in public spaces. IEEE Trans. Intell. Transp. Syst. (ISSN: 1558-0016) 22 (12), 7531–7546. http://dx.doi.org/10.1109/tits.2020.3004000.

Zhang, L., Liu, M., Wu, X., AbouRizk, S.M., 2016. Simulation-based route planning for pedestrian evacuation in metro stations: A case study. Autom. Constr. (ISSN: 0926-5805) 71, 430–442. http://dx.doi.org/10.1016/j.autcon.2016.08.031.

Zhao, Y., Li, M., Lu, X., Tian, L., Yu, Z., Huang, K., Wang, Y., Li, T., 2017. Optimal layout design of obstacles for panic evacuation using differential evolution. Phys. A (ISSN: 0378-4371) 465, 175–194. http://dx.doi.org/10.1016/j.physa.2016.08.021.

Zhao, Y., Lu, T., Fu, L., Wu, P., Li, M., 2020. Experimental verification of escape efficiency enhancement by the presence of obstacles. Saf. Sci. (ISSN: 0925-7535) 122, 104517. http://dx.doi.org/10.1016/j.ssci.2019.104517.

Zhao, X., Wang, N., Han, R., Xie, B., Yu, Y., Li, M., Ou, J., 2018. Urban infrastructure safety system based on mobile crowdsensing. Int. J. Disaster Risk Reduct. (ISSN: 2212-4209) 27, 427–438. http://dx.doi.org/10.1016/j.ijdrr.2017.11.004.

Zhou, Z.-X., Liu, K., Wu, P.-Y., Nakanishi, W., Asakura, Y., 2025. A novel data-driven multi-agent pedestrian flow risk assessment framework for avoiding stampede incident. Accid. Anal. Prev. (ISSN: 0001-4575) 222, 108207. http://dx.doi.org/10.1016/j.aap.2025.108207.

Ziemer, V., Seyfried, A., Schadschneider, A., 2016. Congestion dynamics in pedestrian single-file motion. In: Traffic and Granular Flow '15. Springer International Publishing, ISBN: 9783319334820, pp. 89–96. http://dx.doi.org/10.1007/978-3-319-33482-0_12.